RegimeStress

A Benchmark for Measuring When a Trading Strategy Breaks

counterfactual regime resampling, causal regime labelling, and a validated computational shortcut

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The benchmark in eight numbers

124.5	CPU-hours of counterfactual backtesting, across 100 synthetic histories	**0.897**	rank agreement between the expensive and cheap stress tiers on *performance*
0 / 9	target moments the conditional resampler fails	**0.620**	... and on *fragility*, which is the quantity of interest
0.963	agreement between an expensive fragility measurement and a free one	**0.238**	of R^2 bought by duplicate rows we had not noticed
21.8%	of regime labels disagree with what the same estimator says today	**+0.024**	out-of-sample R^2 predicting fragility, on 109 strategies

Benchmark, protocol, corpus and reference implementation:
github.com/Ashwin-aadi/Trivijaya-Quant-Research

Abstract

A backtest is one draw from a distribution of histories that never got run. Indian equity markets since 2015 contain roughly one pandemic crash, one full rate cycle and one structural liquidity shift, so a claim that a strategy is "robust across regimes" rests on a handful of effective observations no matter how many sessions were simulated. **We release RegimeStress: a benchmark, protocol and reference implementation for measuring how much a strategy's performance moves when the market environment is varied, rather than how high it is on the environment that happened.**

What the benchmark provides. A causally-labelled regime series over 1,233 sessions, estimated by expanding-window refits with a forward-only decode so that no label uses information from after the session it describes; **Counterfactual Regime Resampling (CRR)**, a stationary block bootstrap conditioned on regime label that generates synthetic-but-plausible market histories; a moment-validation suite the synthetic paths must pass;

two stress tiers of very different cost; a frozen strategy population with its exclusion criteria fixed *before* any stress path was run; and a fully reproducible pipeline in which every number in this paper is generated from the committed artifacts rather than transcribed.

> **The cheap stress tier reproduces performance well and fragility poorly.** Resampling a strategy's *realised* returns agrees with full counterfactual re-simulation at Spearman 0.897 on mean performance but only 0.620 on fragility ($n = 125$). Fragility is generated substantially by strategies *responding* to the path they are given, and a shortcut that holds the decisions fixed discards the mechanism. We report this as an empirical methodological result: the expensive tier is not optional.

A shortcut that does hold. Not every shortcut fails, and the benchmark distinguishes them by measurement rather than by assertion. Across-regime fragility computed directly from a strategy's persisted return series agrees with its bootstrapped counterpart at Spearman 0.963 ($n = 125$). The expensive 124.5-CPU-hour experiment was run in full and then used to *validate* the free computation rather than being replaced by an assumption about it.

Fragility prediction: a null, and its diagnosis. Whether fragility can be predicted from strategy characteristics alone — turnover, holding period, concentration, factor exposures — is the benchmark's first task. It cannot, at this sample size. Across 109 strategies and 26 features, five models on identical folds reach an out-of-sample R^2 of +0.024 on the primary target. A permutation test over 200 shuffles nonetheless rejects the null for the *ranking* ($p = 0.005$–0.020) while failing to reject it for the *level* on three of four targets. **Strategy characteristics contain statistically detectable ranking information and very weak information about the absolute level of fragility.** We show the limitation is data variance rather than model bias — every model fits its training folds far better than it generalises, and the gap is *widest* for the model with the most capacity — and therefore add no capacity.

Two corrections, reported as findings. Our own first result was better and wrong. Deduplicating the corpus revealed 11 clusters of strategies with identical realised return series; a duplicate in a training fold with its twin in the test fold was worth 0.238 of R^2. A second defect had one feature column duplicating another under a different name. Both are in the benchmark's corrections log, and the duplicate structure ships as a benchmark artifact — it is a fact about generated strategy corpora, not a blemish on ours.

Disposition. RegimeStress is released as infrastructure to extend, not as a solved problem. The environment, the resampler, the protocol and the exclusion rules are independent of whether our fragility predictor worked, and a larger or more diverse corpus can be scored against exactly the same apparatus.

Contributions

1. **The RegimeStress benchmark.** A frozen environment, strategy population, exclusion protocol and evaluation harness for measuring strategy fragility, with every artifact regenerable by a documented command sequence.
2. **Counterfactual Regime Resampling.** A stationary block bootstrap conditioned on regime label, with block length selected by a stated automatic rule, generating synthetic market histories that preserve cross-sectional structure because dates are resampled rather than assets.
3. **Causal regime labelling.** An expanding-window hidden Markov model with a forward-only decode. We report what causality costs: 21.8% of labels disagree with what the same

estimator says at the end of the sample, and we treat that drift as a property to be reported rather than a defect to be smoothed away.

4. **Validation of the synthetic histories.** 9 target moments, compared against the real series' position in the synthetic distribution. The conditional resampler fails 0; the unconditional one fails 1. We report the failure rather than modifying the resampler until it passes.

5. **An empirical result on when computational shortcuts are safe.** One shortcut is validated at 0.963 and adopted; another is measured at 0.620 and rejected. The benchmark's position is that this is an experiment, not a design decision.

6. **A diagnosed null result on fragility prediction**, separating heavy tails, multicollinearity, influential observations, sample size and genuine weakness of relationship, with the bias-versus-variance question settled by measurement.

1 Why fragility needs a benchmark

The question a portfolio manager asks after "is this signal real?" is "when does it stop working?" The standard apparatus answers the first question much better than the second. Cross-validation with purging and embargo (López de Prado, 2018), deflated performance statistics (Bailey and López de Prado, 2014), and multiple-testing corrections (Harvey et al., 2016) all address selection over *strategies*. None addresses selection over *histories*. A strategy evaluated on 2020–2024 Indian equities has been tested against one sequence of events, and no amount of statistical correction applied within that sequence tells you what would have happened had the sequence been different.

Regime-switching models are the usual response (Hamilton, 1989; Ang and Timmermann, 2012): label the market into states, evaluate within each. This helps, and it introduces two problems that are rarely reported. The first is that regime labels are themselves *fitted*, and fitting them on the full sample before evaluating strategies inside them is a lookahead bias structurally identical to the ones a careful backtest is built to prevent. The second is that the number of independent regime observations is small — far smaller than the number of sessions — so per-regime performance estimates are noisy in a way that per-session counts conceal.

RegimeStress addresses both, and makes the addressing measurable. Our position is that a benchmark in this area is more useful than another method, for the same reason that shared benchmarks proved more durable than individual systems in language understanding (Wang et al., 2019) and code generation (Chen et al., 2021): the apparatus outlives whichever model was state of the art when it was built.

1.1 What this paper is not

It is not a demonstration that fragility can be predicted. Our fragility predictor produces a null result, reported in §3.4 with its diagnosis. It is not a claim that our synthetic histories are the true distribution of counterfactual markets — §4 says at length why they are not. And it is not a strategy paper: no strategy in the corpus is recommended, and many of them lose money.

2 The benchmark

2.1 Population, and exclusions frozen in advance

The input population is two groups kept distinguishable at every stage: strategies generated by a local language model and audited in prior work, and a panel of standard academic factor strategies that acts as a positive control. Pooling them would hide whichever effect is real.

Filter	Count
Strategies entering calibration	185
Failed to execute at all	2
Excluded: not deterministic functions of their inputs	27
of which standard academic factors	0
Stressed	156
Excluded: knife-edge (§3.3)	31
Excluded: exact duplicates (§3.5)	16
In the fragility-prediction training set	**109**

Nondeterminism. 27 strategies return a different result when run twice on the same data with the same code, with a worst-case Sharpe swing of 1.561. Every one is generator output; every standard factor is deterministic. The census was frozen before any stress path was run, which matters: a determinism filter chosen after seeing stress results is a selection rule masquerading as hygiene.

2.2 Causal regime labelling

Regimes come from a Gaussian hidden Markov model (Rabiner, 1989) on realised-volatility features. Three properties are enforced structurally rather than by convention.

The state count is selected once and frozen. $K = 4$, chosen by BIC (Schwarz, 1978) on 1,207 sessions ending 2019-12-31, entirely before the evaluation window. Later refits re-estimate parameters and may never re-select K; the selection artifact is write-once and the script exits with an error if run again.

Parameters are re-estimated on an expanding window, at each quarterly rebalance date, on sessions strictly prior to it. The first refit sees 1,207 sessions and the last 2,378.

The decode is forward-only. This is the part most easily got wrong. Standard HMM libraries offer Viterbi decoding and smoothed posteriors, and *both* decide the state at session t using observations after t. Neither is admissible here. Labels come from $\arg\max_k P(\text{state}_t \mid \text{obs}_{1:t})$ — what an observer standing at t could have known. The cost is a lag of about two sessions at turning points.

What causality costs, reported rather than hidden

21.8% of labels disagree with what the same estimator assigns at the end of the sample, and 9.5% are revised within one quarter. The disagreement is strongly concentrated: calm later years entered the expanding window and recalibrated what "normal" volatility means, so earlier sessions read as more elevated in hindsight.

Consequence, binding on every result in this paper: "state 2 in 2022" and "state 2 in 2024" are not the same conditioning event. Per-period summaries are primary; pooled summaries carry this note. A full-sample fit would have made the labels stable and the analysis invalid.

The regime series is independently anchored: the index closes it is fitted on were compared against the exchange's own published index files on 55 sessions, with 0 beyond tolerance and a worst relative difference of 4.7×10^{-8}. The benchmark also ships a hand-curated timeline of 18 dated market-structure events, each carrying an evidence grade so a reader can see which entries rest on a regulatory circular and which on secondary reporting.

2.3 Counterfactual Regime Resampling

CRR generates synthetic market histories by stationary block bootstrap (Politis and Romano, 1994), with block length selected by the automatic rule of Politis and White (2004) as corrected by Patton et al. (2009): 11.07 sessions at a bandwidth of 14. Two design choices carry the method.

Dates are resampled, not assets. A block is a contiguous run of *sessions*, carried across the whole cross-section at once. This preserves the correlation structure between stocks exactly, at the cost of preserving it *too* exactly — see §4.

Blocks are drawn conditional on regime label, so a synthetic history is assembled from blocks that resemble the regime they are placed into. This is what makes the moments pass.

2.4 Validating the synthetic histories

A resampler that produces implausible histories produces meaningless stress results, so the benchmark requires the synthetic paths to reproduce the real series' distributional properties before any strategy is run on them. 9 statistics are compared — mean, dispersion, skew, tail behaviour, and volatility clustering at several horizons — by asking where the real value falls in the distribution of 1,000 synthetic values over 2020-01-02–2024-12-31.

Resampler	Moments tested	Outside the 95% interval
Conditional on regime	9	0
Unconditional	9	1

The unconditional resampler fails a single long-horizon volatility-clustering statistic, with the real value at the 97.8th percentile of the synthetic distribution.

The failure is reported, not repaired. We did not adjust the resampler until the statistic passed. Conditional resampling is primary and unconditional is retained as an ablation, so the difference between them is an experimental result rather than an assertion about which is better. Conditioning turns out to be nearly free in ranking terms — across-path fragility ranks agree at Spearman 0.991 between the two — but it is what makes the moments pass.

2.5 Two tiers, and two definitions of fragility

Fragility is the ratio of the variance of performance across environments to its mean:

$$F(s) = \frac{\mathrm{Var}_e[\pi(s,e)]}{\mathbb{E}_e[\pi(s,e)]}.$$

The benchmark computes it two ways, because e can range over two different things. **Across regimes**, e indexes the regime labels within the one real history. **Across paths**, e indexes synthetic histories. These are not interchangeable: they agree at Spearman 0.550 ($n = 125$), and both are reported.

The two *tiers* are two ways of obtaining $\pi(s, e)$ on a synthetic path:

- **Tier 1 (faithful).** Rebuild the entire price panel from resampled blocks, then re-run every strategy on it. Strategies re-decide: a momentum strategy on a synthetic path buys whatever that path made a winner. Cost: 124.5 CPU-hours for 100 paths, a median of 4617 seconds per path.
- **Tier 2 (cheap).** Bootstrap each strategy's *realised* return series directly. The strategy's decisions are fixed; only their order is varied. Cost: 23.1 seconds for 1,000 paths.

The ratio in cost is roughly four orders of magnitude, which is exactly why the comparison in §3.1 is worth making.

2.6 Reproducibility, and how every number here was produced

Seed 42 throughout; every stochastic operation takes an explicit seed. Every script writes a run manifest recording git SHA, package versions, input hashes, model tags and wall-clock duration. All inference is local.

No figure in this paper was typed by hand. A generator reads the frozen artifacts and emits every quantitative claim as a LaTeX macro, with the artifact each value came from recorded alongside it; the paper body contains macros only. This is deliberately stricter than our earlier work, where numbers were transcribed from committed reports — correct in practice, but a transcription error would have been indistinguishable from a result.

3 Results

3.1 Tier 2 reproduces performance, not fragility

The central methodological experiment: does the cheap tier substitute for the expensive one?

Quantity	Spearman, Tier 1 vs Tier 2	n
Mean performance	0.897	125
Fragility	**0.620**	125

> **The answer is no, and the shape of the failure is informative.** Tier 2 ranks strategies by performance almost as well as full re-simulation (0.897) and ranks them by fragility considerably worse (0.620). Median fragility differs as well: 0.360 under Tier 1 against 0.311 under Tier 2.
> The interpretation is mechanical rather than statistical. Tier 2 varies the *order* of outcomes a strategy already produced; Tier 1 varies the *world* and lets the strategy respond. That a shortcut preserving decisions loses most of the fragility signal is direct evidence that fragility is generated substantially by strategies responding to conditions — which is what one would hope, and what makes the expensive tier necessary.

We expected Tier 2 to be an adequate approximation and built it as the default. It is not, and the benchmark reports it as an entry rather than removing the tier. Tier 2 remains useful for the question it does answer well, and it is retained so that future work can measure how the gap behaves on a different corpus.

3.2 A shortcut that does hold

Across-regime fragility can be computed directly from a strategy's persisted return series at no simulation cost. The question is whether the free computation preserves the ranking the expensive one produces.

Quantity	Value
Spearman, direct computation vs bootstrap ($n = 125$)	**0.963**
Median fragility, direct	0.618
Median fragility, bootstrap	0.551
Rank convergence at 10 paths	0.817
Rank convergence at 500 paths	0.984

The expensive Tier 1 stress experiment served two purposes: (1) measuring across-path fragility under genuine strategy re-optimisation on synthetic market histories, and (2) empirically validating that the inexpensive computation of across-regime fragility from persisted real return series produces essentially the same ranking (Spearman $\approx$ 0.963), eliminating the need for an otherwise redundant rerun.

The ordering matters. The costly experiment was run first, in full, and then used to establish that a rerun would reproduce a ranking already available for free. It was not skipped and justified afterwards. Taken together with §3.1, the benchmark's position on computational shortcuts is that whether one is safe is an empirical question with a measurable answer, and that the answer differs between shortcuts that look equally reasonable beforehand.

3.3 What the exclusions remove

A strategy is *knife-edge* if its Sharpe ratio is not stable under a numerically negligible change to its inputs — a relative perturbation on the order of 10^{-15}, which no economic interpretation survives. 31 of 156 qualify, including 1 standard academic factor. The rule was applied without exemption.

Median characteristic	Excluded	Retained
Count	31	125
Turnover per session	0.887	0.044
Holding period (sessions)	2.0	37.1
Effective holdings	5.000	5.000

The exclusion is not random, and that is a limitation of our results

The excluded strategies are systematically the *fast* end of the corpus: twenty times the turnover and a twentieth of the holding period of those retained. Every result downstream is therefore conditioned on a restricted range of exactly the characteristic most likely to drive fragility. Sharpe swing among the excluded has a median of 0.0088 and a maximum of 2.514, with 4 above 0.5 — so for most of them the instability is small, and for a few it is not.

3.4 Fragility prediction, and its diagnosis

The benchmark's first task: predict a strategy's fragility from its characteristics alone — turnover, holding period, concentration, book-similarity decay, and factor exposures obtained from one joint regression on all factors at once — without running the stress suite.

Model	R^2	**train** R^2	**gap**	ρ	**MAE**	R^2 **drop-5**
Primary target: fragility across synthetic paths, raw						baseline MAE 2.424
Ridge	−1.742	+0.404	2.146	+0.275	4.887	−271.396
Lasso	−2.259	+0.424	2.683	+0.220	5.620	−343.653
Elastic net	−1.961	+0.415	2.375	+0.250	5.150	−302.146
Random forest	−0.036	+0.292	0.329	+0.225	2.544	−22.049
Gradient boosting	+0.024	+0.653	0.629	+0.108	3.107	−35.864
Fragility across regimes, $\log(1+x)$ *— the only specification that holds together*						baseline MAE 0.422
Ridge	+0.039	+0.459	0.420	+0.405	0.409	−0.595
Lasso	+0.075	+0.433	0.358	+0.340	0.399	−0.444
Elastic net	+0.054	+0.447	0.393	+0.348	0.407	−0.536
Random forest	**+0.212**	+0.537	0.325	+0.320	**0.350**	**−0.002**
Gradient boosting	+0.198	+0.734	0.537	+0.331	0.378	−0.135

Table 1. Five models, identical folds, identical seed, n = 109. *gap* is train R^2 minus out-of-fold R^2. *drop-5* recomputes R^2 after removing the five largest targets, without refitting. All four target specifications appear in the benchmark's `RESULTS.md`; two are shown here.

Is there any signal at all? A permutation test over 200 shuffles separates this from the question of how well it is fit. The observed rank correlation exceeds the null's 95th percentile on all four target specifications, at p = 0.005–0.020. The observed R^2 does not: p = 0.861, 0.507 and 0.572 on three targets, reaching significance only on the fourth (0.005).

> **Strategy characteristics contain statistically detectable information about the fragility *ranking*, and very weak information about its *level*.** That is the benchmark's first reported result, and it is a null with respect to the useful version of the question.

Why, in five parts. A null is only worth publishing if it is diagnosed, so we separated the candidate causes rather than stopping at "the model does not work".

Bias or variance? This decides whether a larger model is warranted, so it was settled by measurement before any was considered. Every model fits its training folds better than it generalises, and the gap is *widest* for the highest-capacity model: gradient boosting reaches +0.734 in sample against +0.198 out of it, a gap of 0.537, while the conservative random forest — the lowest-capacity tree model — attains both the best out-of-fold R^2 (+0.212) and by far the best trimming stability (−0.002 at drop-5 against −0.135).

> **Nothing underfits.** The limitation is data variance, not model bias. We therefore added no capacity: no larger ensemble, no gradient-boosting tuning, no neural network. The binding constraints are 109 rows, a target whose variance sits in five of them, and

Cause	Verdict	Evidence
Heavy-tailed target	**Dominant**	The five largest of 109 rows own 96.1% of the total sum of squares (skew 5.72, excess kurtosis 32.1). An R^2 on this target is arithmetically a statement about five strategies. A log transform reduces the share to 79.1%.
Multicollinearity	**Secondary**	Condition number 253.2; `mean_herfindahl` and `mean_largest_weight_share` correlate at $\lvert r \rvert = 0.995$. It destabilises the linear models severely and leaves the trees alone, so it is not what limits the ranking.
Influential observations	**Confirmed**	Removing one strategy (`candidate_116`) moves out-of-fold R^2 by -1.859, at fixed fold assignment.
Insufficient sample	**Contributor**	The learning curve is still climbing at the largest size available: ρ +0.210 at n = 40 rising to +0.408 at n = 100. It has not saturated.
Genuinely weak relationship	**Partly rejected**	Permutation test as above: real rank information, essentially no level information.

> one shared market history.

Which characteristics matter. After deduplication and the switch to joint factor regression, the leading permutation importances are `beta_long_term_reversal_756d` (+0.408), `factor_r_squared` (+0.280), `trading_session_rate` (+0.187) and `mean_holding_period` (+0.176). Structural characteristics — how often a strategy trades, how long it holds — displaced the factor-exposure fingerprints that led the earlier, contaminated ranking. Given the weakness of the underlying relationship, we present this ordering as descriptive and do not build an argument on it.

3.5 Two corrections, and why they are in the results

Our first predictor result was better than the one above and was wrong. We report the sequence because the correction is more informative than either number.

C1 — duplicate leakage

The generated corpus contains 11 clusters of strategies whose realised net return series are *identical* on every session, to within 10^{-12}: different source code, different stated rationale, same behaviour. 27 strategies sit in a cluster, the largest containing 4; 16 were removed.

Cross-validation assumes exchangeable rows. A duplicate sat in a training fold while its twin was scored in the test fold. Same features, same folds, same seed, differing only in whether duplicates are present: R^2 +0.262 at n = 125 against +0.024 at n = 109. **The leakage was worth 0.238 of R^2**, and it was the difference between a positive headline and a null.

20 further pairs correlating above 0.9999 were detected and deliberately *not* removed: any threshold chosen after seeing its effect on the result is the pathology this programme exists to detect.

C2 — a feature column that was another feature column

One feature was another under a different name, correlating at exactly 1.000 and driving the design matrix's condition number to the order of 10^{15} — a numerically singular design handed to a ridge regression as though it were well posed. Found by a collinearity diagnostic on its first smoke run, not by a test.
Together, C1 and C2 also invalidated the feature importances we had previously reported: the top feature vanished entirely and every magnitude fell by roughly a factor of three.

Duplicate structure ships with the benchmark as an artifact rather than being cleaned away. It is a property of language-model strategy corpora — one candidate in the corpus turned out to be identical in realised returns to the equal-weight baseline it was supposed to beat — and a future generator can be scored on how much of its output is redundant.

3.6 Stress narratives

A local model (`qwen2.5:7b-instruct-q4_K_M`, temperature 0, seed 42) generates natural-language explanations of the most and least fragile strategies, from a prompt containing only measured facts, which are stored alongside each narrative so any claim can be audited.

The weakest component, presented as illustrative

12 narratives were generated at 5.5 seconds each. They are factually correct and near-vacuous: they restate the fragility measurement rather than explain it, and none used the turnover, holding-period or concentration facts supplied to them. One describes a strategy as most reliable in a regime in which it loses money — correct on the measurement, since fragility scores consistency rather than quality, and unhelpful to a reader.
We report this rather than iterating on prompts until the output reads better, which would tune an explainability layer against our own impression of it.

4 Threats to validity

These are the reasons a reader should discount what is above. They are a section rather than a footnote because several of them are load-bearing.

T1 — The synthetic histories are an assumption, not evidence. Across-path fragility rests on 100 counterfactual panels that never happened. The resampler reproduces 9 of 9 target moments under conditional resampling, but reproducing marginal moments is a far weaker claim than reproducing the data-generating process. Every conclusion resting on across-path fragility inherits this in full.

T2 — Conditional resampling does not preserve regime transitions. A bootstrap jump never changes label, so a synthetic path can dwell in one regime far longer than history ever did. Regime persistence in the synthetic data is an artefact of the block length (11.07 sessions), not of the market. The distortion is documented and unmeasured.

T3 — The regime labels are fitted, and the resampler conditions on them. Labels come from an HMM fitted on the same price series the strategies trade, and CRR conditions on those labels. The circularity is real and we have not quantified it. Separately, the expanding-window design means label definitions drift over the sample (21.8% terminal disagreement), so a "regime" is not a fixed conditioning event.

T4 — Excluding nondeterministic strategies removes a non-random group. All 27 excluded strategies are generator output and none are standard factors. The benchmark therefore measures fragility on the subset of machine-generated strategies that happen to be reproducible, and reproducibility may correlate with the simplicity that also drives fragility.

T5 — Excluding knife-edge strategies removes the fast end of the corpus. As §3.3 shows, the 31 excluded strategies have roughly twenty times the turnover and a twentieth of the holding period of those retained. The predictor is trained on a restricted range of the characteristic most likely to matter.

T6 — Tier 2 does not measure what it was built to measure. It is retained in the benchmark and reported honestly, but any future use of it as a fragility proxy is unsupported by our own evidence (§3.1).

T7 — The predictor's sample is small, and its rows are not independent. 109 strategies, summarised over the same market window. There is no time axis to purge — rows are strategies, not periods — but the cross-validation still treats as independent observations that share one realisation of the market. Every confidence interval here is narrower than the truth. The learning curve has not saturated.

T8 — Fragility is a ratio with a mean in its denominator. Strategies whose mean performance is near zero carry a large fragility for an arithmetic rather than an economic reason. 3 are flagged and reported rather than dropped, and how many sit close enough to the boundary for the flag to miss them is unknown. Relatedly, **a uniformly bad strategy scores as stable**: low fragility is consistency, not quality, and nothing in the benchmark enforces that distinction.

T9 — The signal-decay characteristic is a proxy. The engine stores the book a strategy produced, not the alpha forecast behind it, so book-similarity decay measures how fast the realised portfolio turns over. A strategy with a fast signal and a slow rebalancing rule looks slow.

T10 — External validity: one market, one asset class, one period, one generator. Every result is Indian equities, daily bars, a five-year window, and a corpus written largely by a single 7B local model. Nothing here establishes that the tier comparison, the shortcut validation or the prediction null would replicate on US equities, on intraday horizons, or on strategies written by people.

5 Future research

RegimeStress is released as infrastructure to extend rather than as a completed solution. The following are open, and the benchmark is built so that each can be attempted without rebuilding the apparatus.

- **Larger and more diverse strategy corpora.** The single strongest intervention our own diagnosis supports. The learning curve suggests that larger strategy corpora may substantially improve predictive performance; we leave this as future work. A corpus from several generators would also separate "fragility is hard to predict" from "fragility is hard to predict *for strategies this generator writes*".

- **Additional markets and asset classes.** The apparatus is not specific to India. A market with a different participant mix would test whether the tier gap in §3.1 is a property of stress testing or of this market's dynamics.
- **Richer regime models.** Ours labels on volatility features alone. Liquidity, macro state and market-structure epochs are all shipped with the benchmark and unused by the labeller. Regime-switching models with duration dependence would also address T2 directly.
- **Improved fragility predictors with larger datasets.** We deliberately did not add model capacity, because the evidence pointed at variance. That conclusion is conditional on 109 rows and should be revisited at ten times the sample.
- **Alternative stress-generation methods.** Block bootstrap is one choice among several. Generative models of price paths, regime-switching simulators with estimated transition matrices, and historical-analogue methods would each fail differently, and CRR's moment suite is a ready yardstick for comparing them.
- **Extension to intraday strategies.** Daily bars bound what can be said about fast strategies — which is also, uncomfortably, the group our knife-edge filter removes.
- **Evaluating future LLM strategy generators against RegimeStress.** The frozen population, exclusion protocol and fragility definitions give a stable target. A stronger generator can be scored on the same apparatus, including on the duplicate structure of its output, which our corpus turned out to have in quantity.

6 Conclusion

We set out to measure when trading strategies break, and the durable output is the measuring apparatus rather than any strategy-level finding.

Three results are worth restating. Regime labelling can be done causally, and we report what it costs rather than adopting the full-sample fit that would have made the labels look stable. Computational shortcuts are an empirical question: one we tested was validated at Spearman 0.963 and adopted, and another that looked equally reasonable reproduces fragility at only 0.620 and was rejected — by the same experiment, run once, on the same corpus. And fragility prediction from strategy characteristics yields a diagnosed null: detectable ranking information, essentially no level information, limited by data variance rather than model capacity.

Two of our own errors are in the results rather than in an appendix. Deduplicating the corpus removed 0.238 of R^2 that we had reported, and a duplicated feature column had made the design matrix numerically singular. We think a benchmark that shows its corrections is more useful than one that does not, and that the sequence — optimistic result, robustness check, falsification, diagnosis — is the part of this work most worth reusing.

What we release is therefore an environment, a resampler with a validation suite, two stress tiers with a measured account of when each is adequate, a frozen population with exclusion rules fixed in advance, and a pipeline in which every number in this paper is generated from the artifacts rather than transcribed. All of it is independent of whether our fragility predictor worked, which is fortunate, because it did not. It is available at `github.com/Ashwin-aadi/Trivijaya-Quant-Research`.

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